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# IaaC — Infrastructure and Account Continuity
## Executive Brief

### 1. Context

Most organizations on AWS have solved **data backup** reasonably well.
Very few can say, with a straight face,
that they can quickly rebuild their **entire AWS environment**
— VPCs, subnets, security groups, IAM roles, KMS keys, load balancers,
dependencies, and cross-account/region wiring — in a different account or region.

What exists today is fragmented:

- Snapshots and backups (EBS, RDS, S3, AWS Backup),
- Infrastructure-as-Code templates (CloudFormation/CDK/Terraform),
- Years of manual changes, console clicks, and “temporary” hotfixes.

The result:
Data is safe, but **environment continuity** is not.

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### 2. What IaaC Is

**IaaC (Infrastructure and Account Continuity)** is a SaaS concept running on AWS that:

- **Discovers and maps** the *actual* state of an AWS environment (not just templates).
- Builds a **versioned “environment manifest”** describing resources and their dependencies.
- **Automates full-stack recovery** after incidents.
- Acts as a **deployment engine** to clone that environment into **new accounts and regions**.

Think of it as a **DR + deployment layer** for entire AWS accounts, not just their data.

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36 ### 3. Core Capabilities
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38 #### 3.1 Full-Stack Environment Discovery
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40 IaaC inspects an AWS account across:
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42 - **Identity & Access** - IAM users, roles, policies,
   instance profiles.
43 - **Networking** - VPCs, subnets, route tables, gateways,
   security groups.
44 - **Compute** - EC2, autoscaling, containers/functions (ECS
   /Lambda/ECR as needed).
45 - **Storage** - EBS volumes/snapshots, S3 buckets, key
   configuration.
46 - **Control Plane** - CloudFormation stacks, tags,
   CloudWatch metrics/logs.
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48 It doesn't just list resources. It builds a **graph of
   dependencies**:
49 who talks to whom, under which role, in which subnet, using
    which key.
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51 #### 3.2 Environment Manifest & Evidence
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53 From that discovery, IaaC generates a **structured manifest
   **, versioned over time:
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55 - Topology and inventory,
56 - Configurations and metadata,
57 - Backup/DR coverage where applicable.
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59 This manifest serves two purposes:
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61 - **Automation artifact** — input for recovery and
   deployment workflows.
62 - **Evidence artifact** — input for audits and compliance (
   DR plans, tests, RPO/RTO).
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64 #### 3.3 Full-Stack Recovery
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66 Using the manifest plus existing AWS resources (snapshots,
   templates), IaaC can orchestrate:
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 68 - Re-creation of foundational networking and security
    layers,
 69 - Rebuild or relocation of compute workloads,
 70 - Restoration of data from EBS/S3/RDS,
 71 - Reattachment of load balancers and DNS as needed.
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 73 Targets:
 74
 75 - Same account, **new region** (regional DR),
 76 - **New account** (migration, isolation, M&A carve-outs
    ),
 77 - **Clean-room** environments (forensics, testing, audit
    rehearsals).
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 79 #### 3.4 Deployment Engine for New Accounts & Regions
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 81 Once an environment is captured, IaaC can **replicate it
    by design**:
 82
 83 - **Per-customer accounts** for SaaS providers,
 84 - **Regulated accounts** with standardized controls,
 85 - Expansion into **new AWS regions** without a fresh
    manual project.
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 87 Effectively, it turns a working AWS environment into a **
    reusable product**:
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 89 > “Given this manifest, stand up the same platform here,
    under these guardrails.”
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 93 ### 4. Who Benefits
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 95 #### 4.1 SaaS Providers on AWS
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 97 - Faster onboarding for large customers in **dedicated
    accounts**.
 98 - Consistent rollout to **new regions**.
 99 - Stronger DR and compliance story for their own clients.
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101 #### 4.2 AWS Consulting Partners / MSPs
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103 - Rapid **environment assessment** and documentation for
    new clients.
104 - Standardized **DR and continuity offerings** (DR
    rehearsals, DR reports).
105 - Repeatable patterns for **account migrations** and
    modernizations.
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107 #### 4.3 Regulated / Heavily Audited Organizations
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109 - Financial, healthcare, legal, public sector workloads.
110 - Need concrete proof of:
111   - End-to-end recovery capability,
112   - Tested DR procedures,
113   - Controlled deployment to regulated accounts.
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115 #### 4.4 Internal Platform Teams
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117 - Managing **dozens or hundreds of accounts**.
118 - Need visibility into “what’s really running where.”
119 - Need a structured path to **clone** or **separate**
    environments cleanly.
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123 ### 5. Why It Fits the AWS Ecosystem
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125 IaaC doesn’t compete with core AWS services; it **
    amplifies** them:
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127 - Drives more structured use of **EBS snapshots, S3, AWS
    Backup**,
128 - Increases reliance on **CloudFormation, Config,
    Organizations, Control Tower**,
129 - Creates natural hooks for **Step Functions, ECS/EKS,
    DynamoDB, S3** as the control plane.
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131 From a business point of view, IaaC enables:
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133 - DR-as-a-Service and continuity offerings via partners,
134 - Standardized multi-account and multi-region rollout
    patterns,
135 - Stronger alignment with **resilience, governance, and
    well-architected** practices.
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### 6. Roadmap in Brief

After the core discovery → manifest → recovery/deployment loop, the obvious next steps are:

1. **Continuous drift detection**
   Compare current state to prior manifests; flag risky or non-compliant changes.

2. **Risk scoring & recommendations**
   Highlight unprotected volumes, single-AZ dependencies, over-permissive IAM/SGs, etc.

3. **AI-assisted querying of environments**
   Natural-language questions over the manifest graph:
   - “Which workloads are internet-facing without multi-AZ storage?”
   - “What do we need to recreate this environment in `eu-west-1`?”

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### 7. One-Sentence Summary

**IaaC turns a live AWS environment into a versioned, reproducible asset — enabling full-stack recovery, multi-account/region deployment, and audit-ready evidence from a single, AWS-native orchestration layer.**
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